

Documentation

This project builds upon the initial analysis conducted in Project 1, focusing on refining the dataset through sophisticated data cleaning and transformation techniques.

- **Data Sources:**

Details about `historical_stocks.csv` and `historical_stock_prices.csv`.

- **Cleaning Steps:**

- How missing values were handled: forward/backward fill for sector/industry, dropping NaNs after feature engineering
- Outlier detection method (IQR) and resolution strategy (capping).
- Error identification and correction (checks for non-positive prices/volume).

- **Transformation Steps:**

- Feature engineering: daily returns, rolling averages, volatility
- Data standardization: `StandardScaler`
- Encoding of categorical variables: factorize

- **Data Integration:**

`historical_stocks_prices` and `historical_stocks` were **left** merged.

- **Data Splitting:** The methodology for chronological train/validation/test split

We did a 70/15/15 training/validation/test dataset split

- **Assumptions:** We factorized categorical variables to transform or convert them into numerical values on the sector and industry columns of the `historical_stocks`

- **Tools and Libraries:** List all Python libraries used: Pandas, Scikit-learn, Matplotlib, Seaborn